Why Does the Number 3 Appear in $u_{\mathrm{rms}} = \sqrt{3RT/\mathcal{M}}$?

- Question
  - Why does u_{rms} contain the factor *3*?
- Evidence
  - Gas molecules translate independently along three perpendicular directions.
  - Each translational direction contributes equal average kinetic energy by *equipartition*.
- Reasoning
  - $\because$ Three equal contributions add, total translational energy is three times one-direction energy.
  - $\therefore \langle u^2 \rangle = 3RT/\mathcal{M}$, giving $u_{\mathrm{rms}} = \sqrt{3RT/\mathcal{M}}$; fewer dimensions give smaller factors.
- Takeaway
  - The factor 3 counts translational dimensions.